

```
# 1. Import libraries such as Pandas, matplotlib, NumPy, and seaborn and load the insurance dataset
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns

df = pd.read_csv('insurance.csv')
df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Next steps: [Generate code with df](#) [View recommended plots](#) [New interactive sheet](#)

```
# 2. Check the shape of the data along with the data types of the column
# Check the shape of the data
print(f"Shape of the dataframe: {df.shape}")

# Check the data types of the columns
print("\nData types of each column:")
print(df.dtypes)
```

Shape of the dataframe: (1338, 7)

Data types of each column:

```
age      int64
sex      object
bmi      float64
children int64
smoker   object
region   object
charges  float64
dtype: object
```

```
# 3. Check missing values in the dataset and find the appropriate measures to fill in the missing values
# Check for missing values in the dataset
print("Missing values in each column:")
print(df.isnull().sum())
```

##the results show that there are no missing values

Missing values in each column:

```
age      0
sex      0
bmi      0
children 0
smoker   0
region   0
charges  0
dtype: int64
```

Code for Exploratory Data Analysis (EDA): **Count plot for categorical columns:** A count plot is useful for visualizing the frequency distribution of a categorical variable.

Scatter plot for numerical columns: A scatter plot helps visualize the relationship between a numerical feature and the target variable (also numerical).

```
# 4. Explore the relationship between the feature and target column using a count plot of categorical columns and a scatter plot of numer
import matplotlib.pyplot as plt
import seaborn as sns

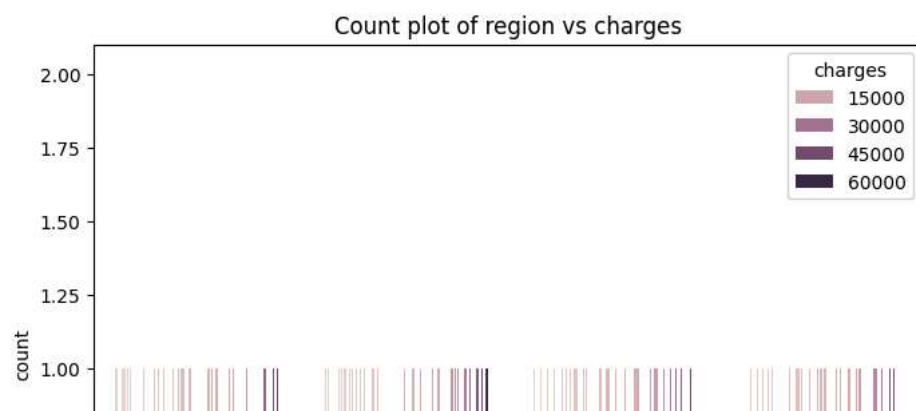
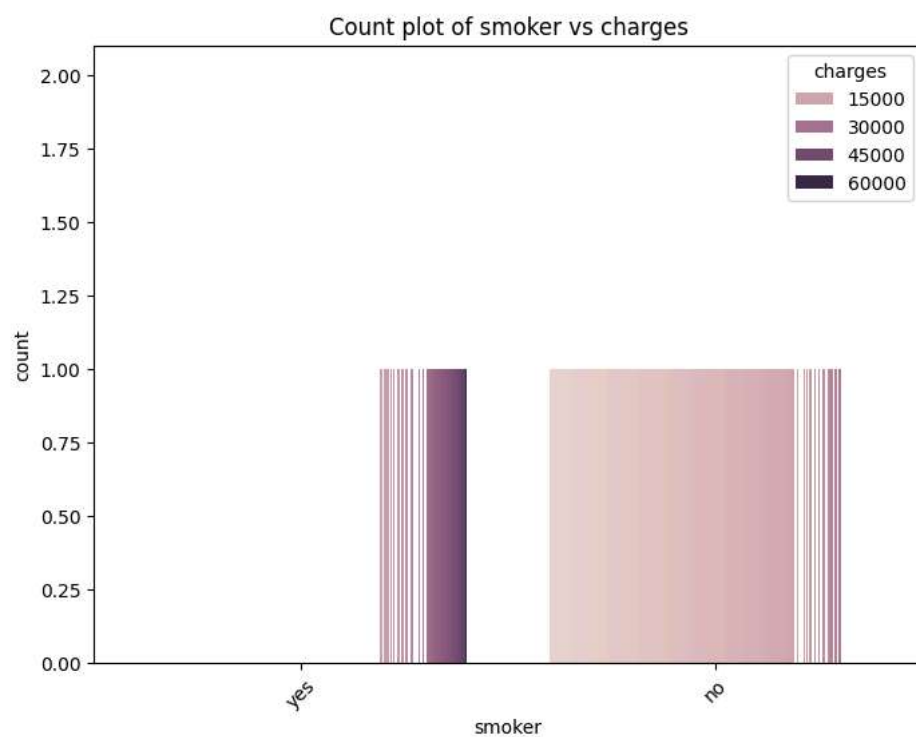
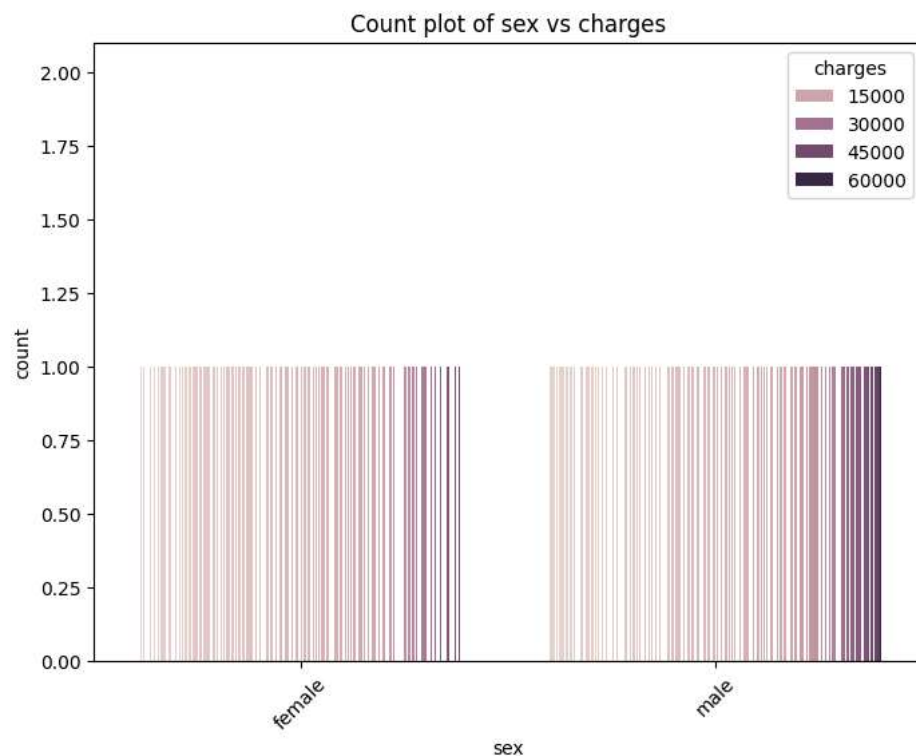
# Set the figure size
plt.figure(figsize=(12, 6))

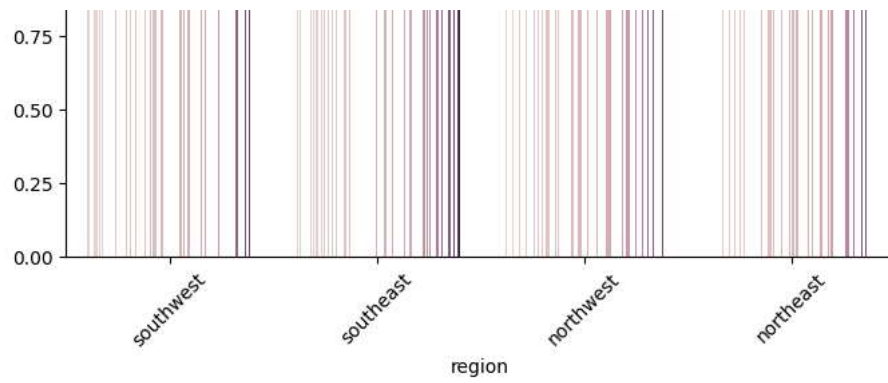
# Count plot for categorical features
categorical_columns = ['sex', 'smoker', 'region']
```

```
for col in categorical_columns:
    plt.figure(figsize=(8, 6))
    sns.countplot(data=df, x=col, hue='charges') # 'Rating' is the target variable
    plt.title(f"Count plot of {col} vs charges")
    plt.xticks(rotation=45)
    plt.show()

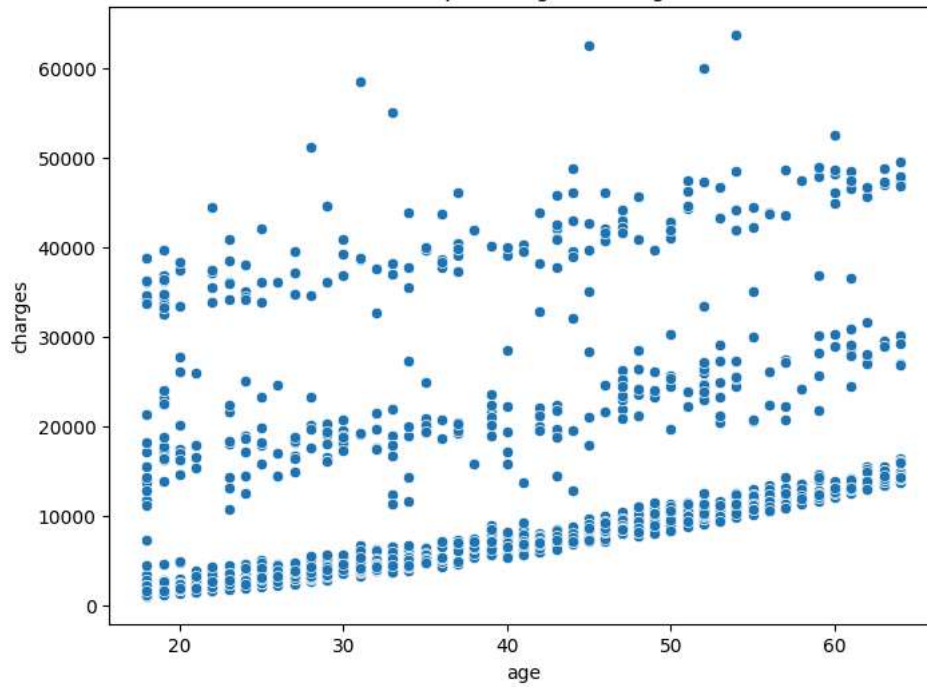
# Scatter plot for numerical features
numerical_columns = ['age', 'bmi', 'children', 'charges']

for col in numerical_columns:
    plt.figure(figsize=(8, 6))
    sns.scatterplot(data=df, x=col, y='charges') # 'Rating' is the target variable
    plt.title(f"Scatter plot of {col} vs charges")
    plt.show()
```

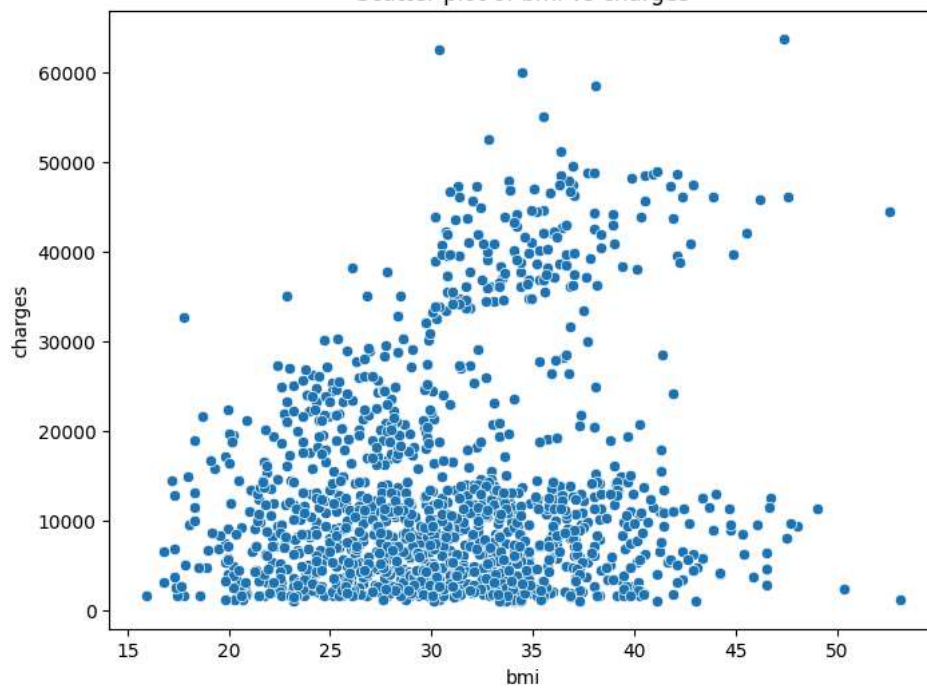
 <Figure size 1200x600 with 0 Axes>



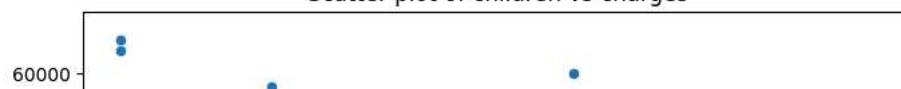
Scatter plot of age vs charges

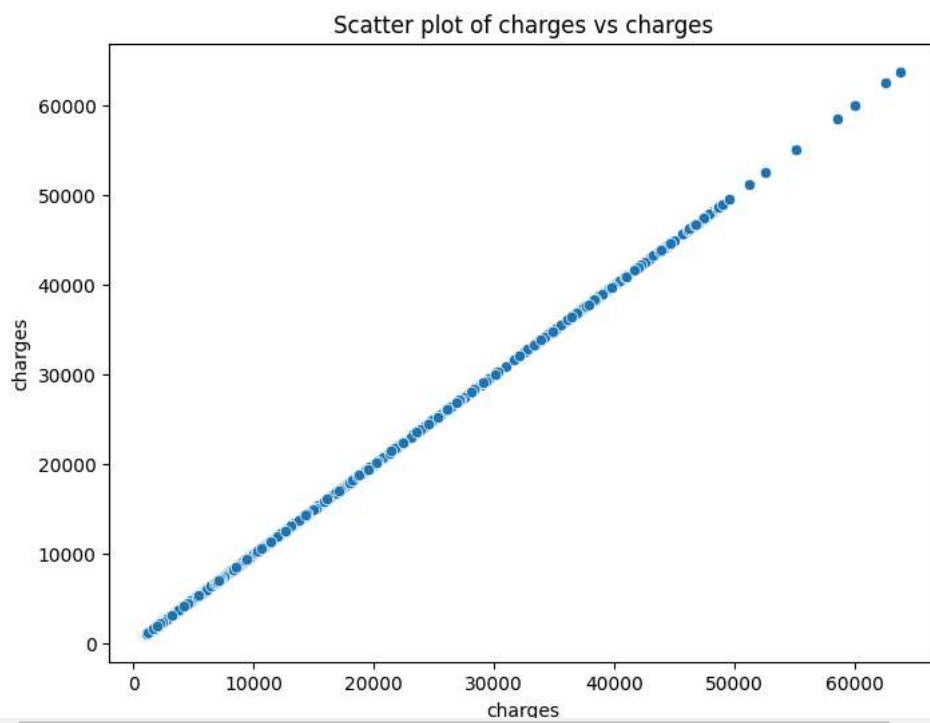
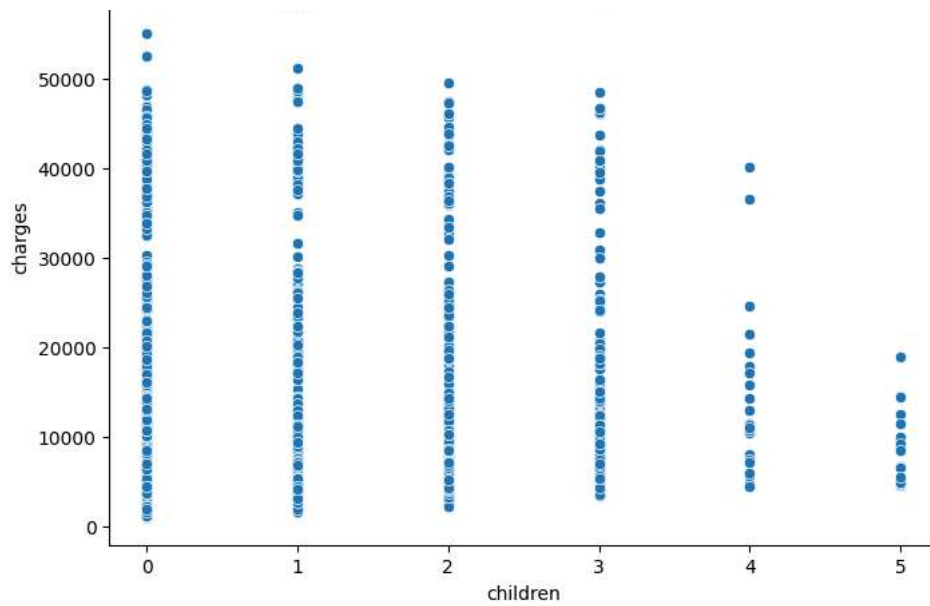


Scatter plot of bmi vs charges



Scatter plot of children vs charges





Visualization

```
# Pairplot to visualize relationships  
sns.pairplot(df, hue="smoker", palette="husl")  
plt.show()
```

